

# Thinking Alone, and Together: Dissenting Pairs Corrected More Faulty Decision Judgments Than Solitary Reasoners Across Four Tasks

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Working paper

**Abstract.** Humans usually reason in groups. Yet contemporary reasoning research relies almost exclusively on people solving problems alone. This failure of ecological validity partly reflects the difficulty of running group-reasoning studies online. In response we developed the Socrates platform to conduct online discussion-based experiments on any topic. In four pre-registered experiments ( $N = 1672$ ) and with a range of canonical reasoning problems, we show that participants who talk with a dissenting peer correct faulty judgments three to four times more frequently than those who reflect alone. Our findings underline the importance of studying reasoning in argumentative contexts—its natural home—and we show how this can be done affordably and at scale. We conclude by exploring the implications of this line of inquiry for our understanding of how human reasoning works, how it should be measured, and how it can be improved.

**Supplements:** [https://osf.io/fp2z4/?view\\_only=006f729be8b4491e9593de35ff651aaa](https://osf.io/fp2z4/?view_only=006f729be8b4491e9593de35ff651aaa)

**Keywords:** reflective equilibrium, critical thinking, belief bias, judgment and decision-making, peer disagreement

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*A psychologist [...] once confessed to me that despite all problems in short-term inferences like the Wason Card Task, there was also the undeniable fact that he had never met an experimental subject who did not understand the logical solution when it was explained to him, and then agreed that it was correct.*

– Johan van Benthem (2008). "Logic and reasoning: do the facts matter?"

## Introduction

From insects and fish to birds and mammals, a remarkable variety of creatures solve problems in groups (Krause et al., 2010). These include coordination problems, such as those involved in hunting large prey (Packer & Ruttan, 1988; Stander, 1992) and—even more remarkably—distributed decisions that integrate many individuals' partial evidence, as when honeybee swarms choose a nest site via quorum-like commitment (Seeley, T.D., 2010).. While each member of a group may face severe cognitive limitations, the group itself is capable of astonishing feats. It should come as no surprise, then, that a group-level intelligence factor may also explain a group's performance beyond its members' average or maximum intelligence (Riedl et al., 2021; Woolley et al., 2010). Investigating human reason by studying solitary

reasoners is therefore like investigating primate grooming behavior by studying solitary monkeys: many will self-groom when they are alone, but to study the most adaptive features of grooming, we must observe it in social contexts.

A rich tradition in psychology has investigated group reasoning and its findings are clear: Like other social animals, humans often solve cognitive challenges far better together than they do alone (Davis, 1969; Hastie, 1986; Hill, 1982; Klein & Epley, 2015; Laughlin et al., 2006; McGrath, 1962). The effect emerges among adults and children working on both classic reasoning problems and real-world tasks (Mercier, 2011; Silver et al., 2021). The benefits of discussion are so impressive that even small samples of group reasoners surprise both laypeople and experts (Mercier et al., 2014). These findings make sense if human reasoning is adapted, at least in part, for social goals—especially argumentation (Mercier & Sperber, 2017). Yet inquiry into this distinctive human capacity relies almost exclusively on people completing surveys alone (Solano, 1987). And unlike the contexts in which human reason evolved where much including our lives can depend on getting it right, in typical studies of reasoning, nothing weighty rides on participants' arriving at good answers (Litman et al., 2023).

These failures of ecological validity are far from the only limitations of standard survey-based methods. A large and compelling literature demonstrates that (1) comprehension errors are common and difficult to detect due to metacognitive limitations (Holbrook et al., 2007); (2) low participant motivation promotes careless, effort-minimizing response styles (Vannette & Krosnick, 2014)—an especially serious problem in studies that involve challenging reasoning problems; and (3) conversational norms license contextual inferences that are invisible to experimenters (Cullen, 2010; Schwarz, 2014), making it unclear whether participants interpret stimuli as intended (Hertwig & Gigerenzer, 1999; Mandel, 2014; Politzer & Macchi, 2000).

While survey methodologists have developed tools to mitigate some of these problems (Byrd et al., 2023; Krosnick et al., 1990; Oppenheimer et al., 2009), the findings presented here suggest that discussion-based studies have distinct advantages (Allport, 1920). People working together to find the best answer to a tricky problem can uncover comprehension errors and unintended pragmatic influences (Bennett et al., 2010). Group-level incentives and the intrinsically motivating context of discussion can motivate (Morrow, 2022) and even depolarize (Combs et al., 2023).

Given the case for studying reasoning in argumentative contexts and a long history of group reasoning research, why is it that discussion-based studies play a tiny role in contemporary psychology of reasoning? Part of the explanation is that running such studies is difficult, costly, and time consuming, and many researchers lack the tools required to implement them on common web-based platforms. Indeed, online studies of group problem solving often rely on nominal groups (Taylor et al., 1958). The members of a nominal group cannot engage in the back-and-forth characteristic of natural discussion. Instead, some produce outputs that are later fed to others isolated from the original producers (Kosinski et al., 2012). While nominal groups outperform interactive groups on some decision-making tasks (Van de Ven & Delbeco, 1971), in the context of reasoning research they are a pale substitute for discussions where arguments and counterarguments are exchanged and refined in real time. Of course, real-world argumentation often occurs in uncontrolled and difficult-to-replicate contexts such as science fairs, conferences, and classrooms (Mercier & Claidière, 2021). Researchers need a better alternative, one that preserves the advantages of recruiting participants online—rapid and affordable data that represents diverse populations—while making it easy to study group reasoning in a controlled, replicable, and more ecologically realistic setting.

## Present Research

In this paper, we introduce *Socrates* – a free and open-source platform for conducting real-time, interactive reasoning experiments online and in the field. Built on oTree (Chen et al., 2016), *Socrates* provides an easy-to-use, no-code environment to design and run large-scale, discussion-based studies. It interfaces with popular online labor markets such as CloudResearch and Prolific Academic, allowing researchers

to recruit larger, more diverse, and more affordable samples than those used in early group-reasoning research.

In a series of four pre-registered studies, we demonstrate that discussions between dissenting peer caused significantly more improvement than deterioration on a diverse set of tasks while reflecting alone did not, suggesting that pure truth-wins models of group problem solving underestimate its value and only partly characterize its mechanisms (Cooper & Kagel, 2005; Friedkin & Bullo, 2017).

Our stimuli included adaptations of the Wason selection task (Wason, 1968), a base rate neglect problem (Tversky & Kahneman, 1977), a belief bias question (Janis & Frick, 1943), and a philosophical thought experiment (Gettier, 1963; Weinberg et al., 2001). To avoid confounds between reflection and test familiarity (Byrd, 2023; Byrd et al., 2023), the wording of each task was modified from the original and then pre-tested (Byrd, 2025b). Figure 1 shows all resulting stimuli.

Most interventions that aim to improve reflection test performance among solitary reasoners produce disappointing results. One of the most successful interventions involves teaching people how to solve reflection tests (Isler & Yilmaz, 2022). Alas, the benefits of such interventions do not generalize to other reflection tests (Calvillo et al., 2023). These findings suggest that improving reflection test performance is not only more difficult than previously anticipated, but also a more circumscribed goal than what we actually care about: helping people think more reflectively about more than just contrived math or logic puzzles.

(Yechiam & Zeif, 2022) We recruited 1825 participants from CloudResearch’s “Approved” Amazon Mechanical Turk workers (Byrd, 2025b; Hauser et al., 2023) for a base fee of \$2.00 USD. Our experimental design has four parts. In the first, participants respond to a reasoning problem (Figure 1, panels A-D) by selecting one of two response alternatives and indicating their confidence. In the second, they complete a test designed to familiarize them with the experimental procedure. One hundred fifty-three participants either failed the test at least three times or dropped out of the study for some other reason, leaving a final sample of 1672 (for demographics, see Methods). In the third part, participants completed the reflection intervention. All participants were told that “the aim of this exercise is to arrive at the best answer” and reminded they could “change” their response to the problem after discussion or essay writing. Participants who gave opposing responses in part one were matched into pairs and redirected to text-only chat rooms to discuss it together for as long as they wished. Participants who could not be matched were redirected to a similar-looking solitary reflection room. To explore the effect of different reward structures, participants were randomized into one of three bonus conditions. In the independent bonus conditions, participants learned they would earn a \$0.50 bonus if their own final answer was the best response. Participants who were paired for discussion could be randomized into the “dependent” bonus condition where they learned that they would earn a bonus only if *both* chat partners’ final responses were best. Finally, in the no bonus conditions, participants were not informed of any opportunity to earn a bonus.

In the final part of the study, participants indicate their final answer and confidence before completing a brief exit survey. Pre-registration, data, scripts, and materials can be found on the Open Science Framework:

<https://osf.io/fp2z4/>

## Results

The tasks reliably lured people away from the best response. Across conditions, just 28% of respondents were correct before reflection, and only 41% were correct afterwards:  $d = 0.27$ ,  $t(1671) = 11.48$ , 95% CI [0.11, 0.15],  $p < 0.001$ . That small improvement was observed on all four tasks ( $p < 0.001$ ) across conditions. However, our primary question is whether participants in the discussion conditions improved meaningfully more than those in the solitary reflection conditions.

Consider the proportion of decisions that improved after reflection, subtracting the proportion of decisions that worsened. Figure 3 illustrates how, across our four stimuli, this *net correction* was three to four times larger when thinking together (25%) than when thinking alone (7%,  $p < 0.001$ ). Thinking together caused more corrections on net ( $p < 0.001$ ); thinking alone did not ( $p > 0.2$ ).

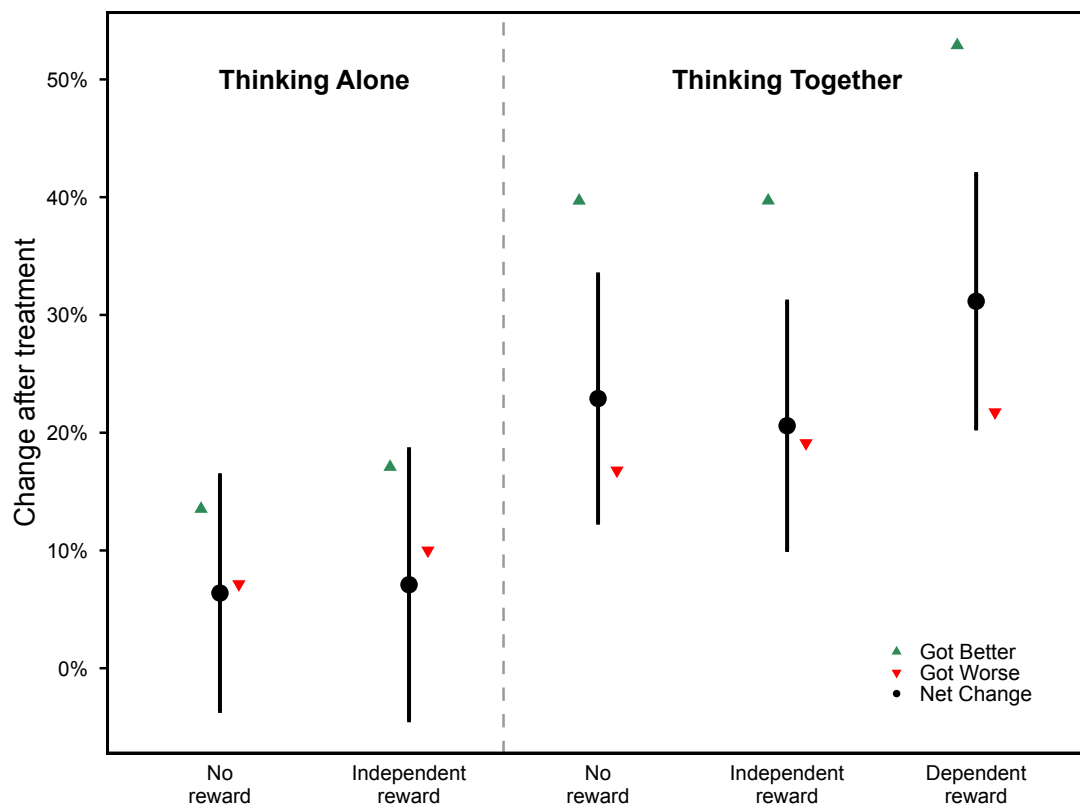
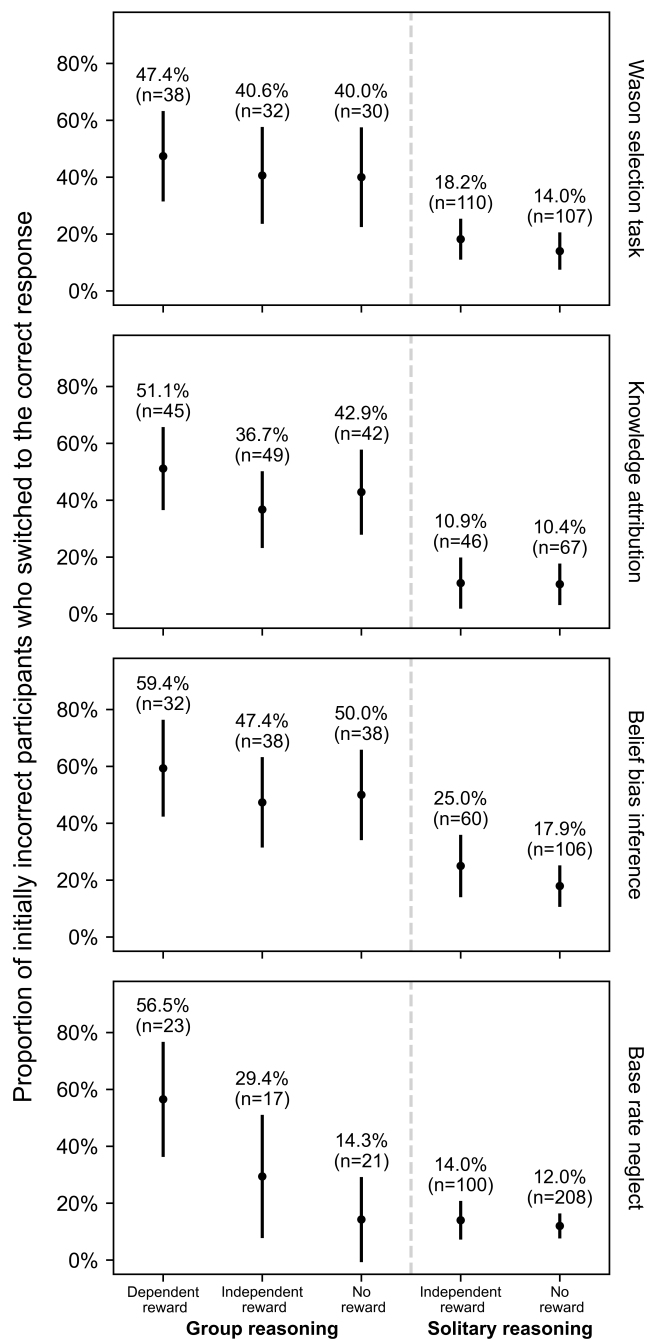


Figure 3. Net correction by condition, aggregating across four tasks. Error bars represent 95% confidence intervals. Thinking alone yielded six to seven percent net correction ( $p > 0.2$ ), but thinking together yielded twenty to thirty percent net correction ( $p < 0.001$ ). The three-to-five-fold advantage of thinking together was significant ( $p < 0.001$ ).



Notably, reflecting together outperformed solitary reflection despite a smaller opportunity for improvement: in the discussion conditions, one participant always initially selected the best response, meaning that at most 50% of participants could correct their initial judgments, while in the solitary reflection conditions, the maximum proportion of corrections could easily exceed 50%.

Also, conversation seemed better than cash. The financial bonus for best final decisions did not seem to make a difference to net correction rates.

Moreover, we observed the benefit of thinking together on both the non-demonstrable philosophy task and the logico-mathematical tasks with demonstrably correct and incorrect response options? ( $p < 0.02$ ). Indeed, Figure 4 visualizes how thinking together caused net correction on both kinds of stimuli ( $p < 0.001$ ), but thinking alone did not ( $p > 0.2$ ).

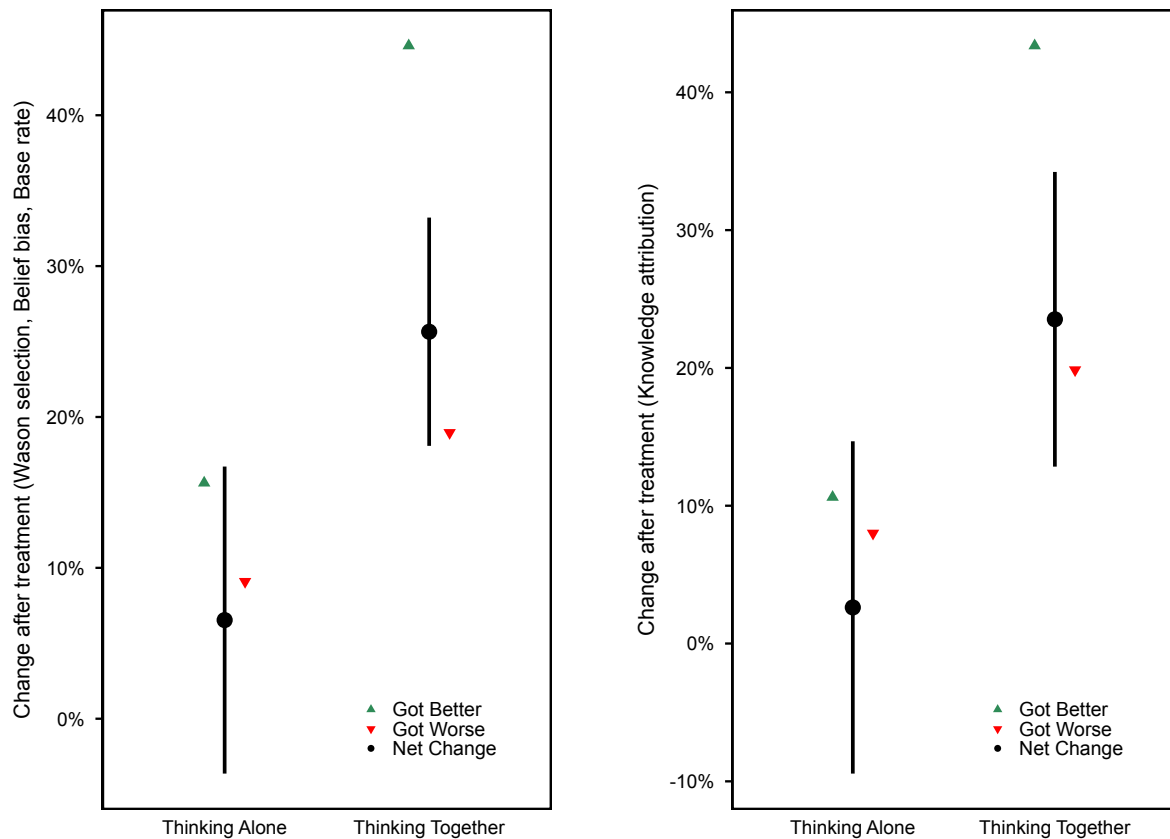


Figure 4. Net correction for demonstrable and non-demonstrable stimuli by togetherness. Error bars represent 95% confidence intervals. Again, thinking alone yielded up to about seven percent net correction ( $p > 0.2$ ), but thinking together yielded between twenty and thirty percent net correction ( $p < 0.001$ ). The advantage of thinking together was detected in both kinds of stimuli ( $p < 0.02$ ).

## Discussion

Reflecting on a problem with a dissenting peer led to the greatest proportion of correcting initially faulty intuitions, far outweighing cases in which decisions became worse. Some evidence such Socratic reflection could yield even more decision improvement face-to-face (Gajendran et al., 2022) or with artificially intelligent chatbots (Boissin et al., 2025).

**Alignment with prior work.** These results provide further promise for discussion that follows independent judgments (Dezechache et al., 2022; Minson et al., 2018). A key contribution of our experiments are controlled comparisons between participants that do and do not encounter discussion or bonus opportunities during reflection. By including these additional conditions in our design, we were able to detect that discussions enhanced reflection above and beyond the potential effect of incentives.

**Disagreement or discussion?** There are open questions about whether the goal of argument is to handle disagreement (Aikin & Casey, 2022). Social exchanges of information seem to improve individuals' judgments even if they reduce group-level accuracy (Becker & Fletcher, 2022). However, merely encountering disagreement may also influence reasoning in ways that leads to "a better decision" (Panggabean et al., 2022, p. 273). And one difference between the discussion conditions and the essay conditions is whether someone encounters disagreement. So there are opportunities to empirically test whether the improvements we observed require discussion and disagreement or just disagreement. In principle, one may be able to exploit "dialectical bootstrapping" by imagining dialogical disagreement instead of actually discussing with a disagreeing peer (Herzog & Hertwig, 2009). One way to test this would be to introduce *disagreement* into the *essay* conditions (e.g., by telling participants that other participants disagreed with

their response). Nonetheless, existing data include discussions between people who *already agree* (as well as those who disagree) and the results suggest that discussion remains helpful regardless (Karadzhov et al., 2022). Additional research could profitably investigate how rationality and other aspects of reasoning vary between these conditions.

**Incentives.** In the aggregate, performance bonuses did not seem to benefit reflection, which aligns with many prior findings (Camerer & Hogarth, 1999; Isler & Yilmaz, 2022; Sirota et al., 2023). However, on the base rate neglect task where few participants were initially correct, participants seemed outstandingly resistant to changing their response. Only when participants were offered a financial bonus that *required* not just the *correct* final answer but also *agreement* between discussants on that correct answer did performance improve compared to the other conditions. This suggests that there are circumstances in which discussion can be fruitfully supplemented by cash incentives. We hypothesize that cash bonuses may make more of a difference when (a) different aspects of the stimulus are more salient to each player and the information from both sides is needed to solve the problem (Zonca et al., 2020) or (b) when the bonus is more meaningful than just 25% of base pay (Yechiam & Zeif, 2022).

**Further use.** This Socrates platform could be fruitfully used beyond basic decision science experiments. For instance, its remote access and scalability could offer “philosophy competitions” to reach more people in more places (Easton, 2022). Intelligence tradecraft has also been theorized to benefit from consideration of “alternative hypotheses [and] mind-sets” (Heuer, 1999, pp. 44–45, 180–181). Even philosophers have proposed that conversational survey designs will enhance our understanding of philosophical thinking (Cullen, 2010; Hansen et al., 2022). To enable interested parties to facilitate and test the effects of Socratic reflection on in their own research programs, the Socrates platform is opensource: [github.com/byrdnick/socrates](https://github.com/byrdnick/socrates).

**Resolving puzzles about reflection.** The interpretation of some decision-making results depend on the degree to which people’s decisions involve careful reflection on all relevant information (Kauppinen, 2007; Kneer et al., 2021). In some of these fields there are mixed results (Byrd, 2025b). Some researchers emphasize similarities across demographics and framings (Knobe, 2021; cf. Stich & Machery, 2022) while other researchers emphasize significant differences (Gonnerman et al., 2022; Machery et al., 2017, 2018; Ziolkowski, 2022). One proposal is that at least some of the differences have more to do with unintended interpretations of vignettes, questions, and/or response options (Cullen, 2010). The Socrates platform was designed in part to help overcome such problems by placing reasoners in an environment where mistaken readings and interpretations are more likely to be noticed and corrected: discussion. Insofar as this is the actual effect of the chats we facilitated, Socrates could be a promising tool for empirically resolving unresolved puzzles about whether and how reflective decisions vary by demographics and framings. If the differences remain when thinking alone and together, then this would favor the similarity hypothesis. If, however, the differences are detected only in when thinking alone, but not when thinking together, this would suggest that the differences were the result of insufficient reflection rather than deep differences between demographics and frames.

## Materials

*Wason selection task* (Fig 1, top left). People often prioritize confirmatory information over falsifying information when testing hypotheses, even when reaching the correct response requires an inference from that information. We adapted the Wason Selection Task to test whether discussion mitigates this tendency (Wason, 1968).

*Belief Bias* (Fig 1, top right). People often evaluate arguments based on the believability of their conclusions. Because logically invalid arguments can have believable conclusions, this tendency generates faulty judgments (Janis & Frick, 1943). We adapted an invalid argument with an appealing conclusion to explore whether discussion helps people overcome this bias.

*Base Rate Neglect* (Fig 1, bottom left). People often ignore base rate information when estimating probabilities. Apart from reframing problems in terms of frequencies, this bias is extremely stubborn. We modified

Tversky and Kahneman’s (1977) problem to test whether discussion helps participants integrate base-rate information into probability estimates. Most people’s estimates ignored the base-rate information, even though a quick calculation can show that this is incorrect (Byrd, 2025a).

*Accidentally True Belief* (Fig 1, bottom right). Philosophers rely on judgments about hypothetical scenarios to analyze the application conditions of many concepts. Early analyses of the concept of *knowledge* held that a person knows that  $p$  (where  $p$  is any proposition) if and only if  $p$  is true and the person is justified in believing  $p$ . Philosophers developed thought experiments (Gettier, 1963, pp. 122–123; Russell, 1948, p. 170) that have culminated in a majority of professional philosophers rejecting that analysis (Bourget & Chalmers, 2023). We adapt our vignette from Weinberg, Nichols, and Stich’s (2001) version of the thought experiment.

Denying knowledge (B) is the orthodox response for such cases of accidentally true belief, usually favored more by relevant experts in philosophy than other philosophers, other academics, or laypeople (e.g., Horvath & Wiegmann, 2016; Starmans & Friedman, 2020). Some experiments attempted to test the effect of reflective thinking on judgments about Gettier cases (Kneer et al., 2021). Alas, these experiments did not report whether the manipulation improved performance on behavioral tests of reflection. So it remained unclear whether the correlations between philosophical expertise and knowledge denial in Gettier cases represents is caused by or merely correlates with reflection.

## Participants

For a margin of error of about  $d = 0.4$ , precision for planning analysis recommends 50 participants per group (Cumming & Calin-Jageman, 2016) in each of the five conditions (50 discussions with a cooperative bonus, 50 discussions with an individual bonus, etc.). Because we cannot control how many participants will be matched, we rely on pilot studies to select stimuli that produce a roughly even distribution (Byrd, 2025b). Across all tasks, 1825 participants were recruited through CloudResearch’s “approved” list of mTurk workers (ibid.) and 153 did not provide complete data or automatically exited after failing the comprehension test three times, leaving a final sample of 1672 ( $M_{\text{age}} = 40$ ; 800 female, 863 male, 6 neither; 1363 White, 138 Black or African American, 114 Asian, 13 American Indian or Alaska Native, 1 Native Hawaiian or Pacific Islander, 40 Other; 662 less than Bachelor’s degree, 745 with Bachelor’s degree, 262 post-bachelor’s degree; more than 1600 born and living in the United States).

## Performance Measures

The Socrates app collects various performance metrics throughout the experiment and responses to a debriefing survey that comes after the main experiment. The most relevant variables in the resulting dataset are as follows.

| Measure                        | Summary                                                      | SD                    |
|--------------------------------|--------------------------------------------------------------|-----------------------|
| Initial decision status        | 28% correct initially                                        | 0.45                  |
| Initial confidence             | 8.22 on a 0–10 scale                                         | 2.04                  |
| Initial decision time          | 49 seconds                                                   | 77                    |
| Comprehension errors           | 0.52                                                         | 0.68                  |
| Final decision bonuses         | 367 of 1672 participants received a bonus; mean 0.11], [0.21 | Final decision status |
| 41% correct after intervention | 0.49                                                         | Final confidence      |
| 8.36 on a 0–10 scale           | 1.96                                                         | Final decision time   |
| 19 seconds                     | 23 seconds                                                   | Discussion duration   |

|                           |             |                        |
|---------------------------|-------------|------------------------|
| 189 seconds               | 130 seconds | Essay duration         |
| 162 seconds               | 149 seconds | Intervention duration  |
| 175 seconds               | 141 seconds | Age                    |
| 40 years                  | 12 years    | Participant experience |
| 4 of 5, somewhat positive | 0.92        |                        |

Dataset variables include first.1.player.answer, first.1.player.confidence, first.1.player.time\_on\_first\_opinion, first.1.player.cq\_err\_counter, second.1.player.payoff, second.1.player.answer, second.1.player.confidence, second.1.player.time\_on\_second\_opinion, second.1.player.time\_on\_discussion, second.1.player.time\_on\_essay, seconds\_on\_intervention, and Questionnaire.1.player.age.

Categorical variables collected during the debriefing survey (including the coding) are included in the table below.

| Variable                   | Coding and counts                                                                                                                                                         | Dataset label                     |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Sex                        | Male 863; female 800; other 6                                                                                                                                             | questionnaire.1.player.sex        |
| Race                       | White 1363; Black or African American 138; Asian 114; American Indian or Alaska Native 13; Native Hawaiian or Pacific Islander 1; other 40                                | questionnaire.1.player.race       |
| Education                  | Less than high school 4; high school 156; some college 318; associate degree 184; bachelor's degree 745; master's degree 215; doctoral degree 29; professional degree 18  | questionnaire.1.player.education  |
| Socioeconomic status       | Less than high school 62; high school 402; some college 213; associate degree 176; bachelor's degree 560; master's degree 191; doctoral degree 28; professional degree 37 | Questionnaire.1.player.ses        |
| Country born               | United States 1602; other 67                                                                                                                                              | Questionnaire.1.player.country_bo |
| Country lived most of life | United States 1648; other 21                                                                                                                                              | Questionnaire.1.player.country_li |

The formula for the CI of a proportion is:

$$CI = \hat{p} \pm z \times \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Where  $p$  is the sample proportion,  $z$  is the z-score corresponding to the desired confidence level (1.96 for 95% CI), and  $n$  is the sample size. The `stats.norm.interval` function is used to calculate this, where `loc` is the mean (the sample proportion in this case) and `scale` is the standard deviation of the sampling distribution of the proportion.

We calculate a CI for the net change in proportions using the standard error for the net change, which is calculated by taking the square root of the sum of the variances of the two proportions. The CI (Wald) is then:

$$CI = (\hat{p}_1 - \hat{p}_2) \pm z \times \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

## Robustness

The observed effect of dissenting discussion on gross correction remains and becomes larger after controlling for other observed variables: Figure A5 shows that thinking together was at least 1000% more likely to correct initial responses, on average, compared to thinking alone without a bonus ( $p < 0.001$ ). The cash bonus on its own (in solitary reflection) predicted just 44% higher odds of correction, on average.

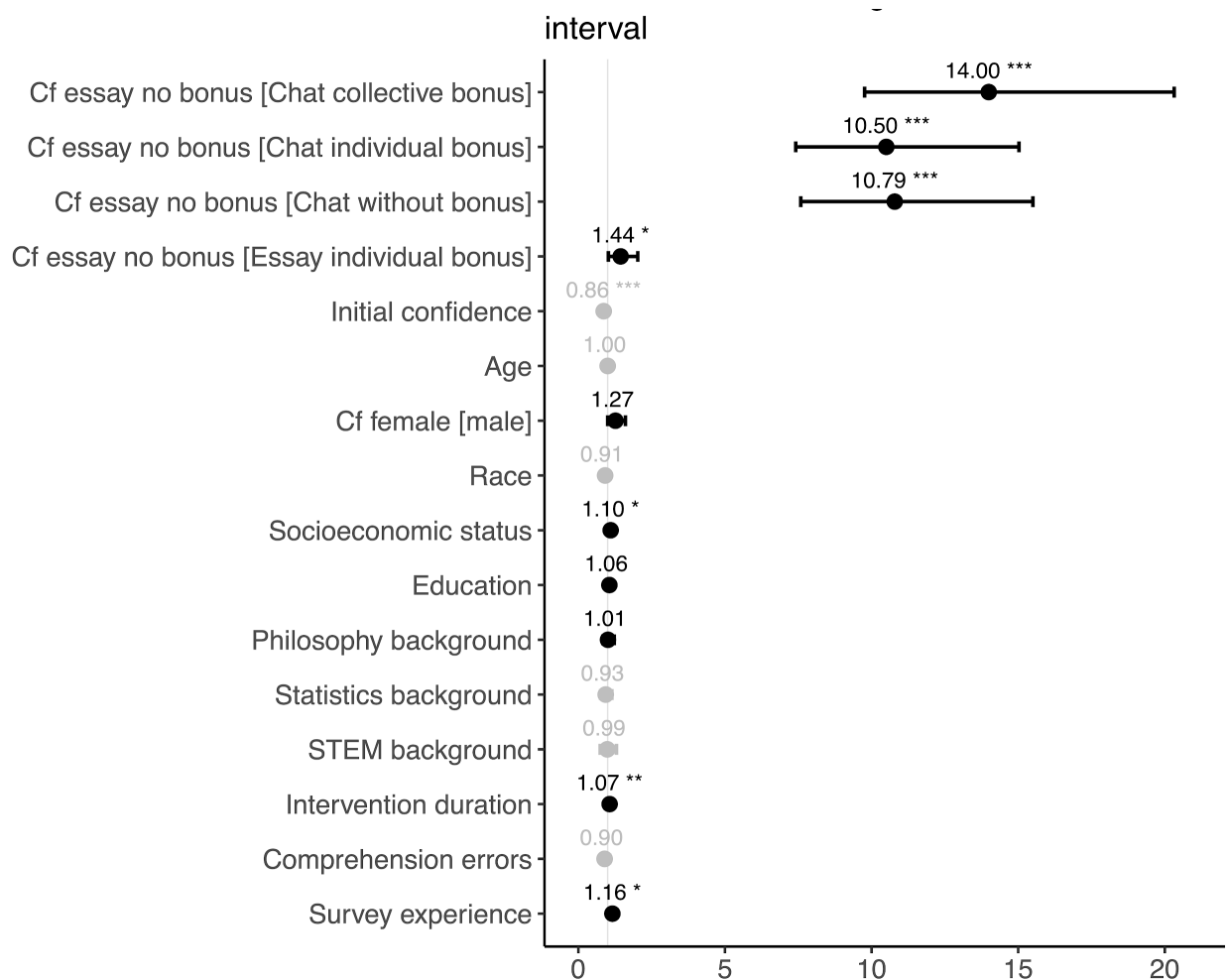


Figure A5. Odds of gross correction with 95% confidence intervals. Cf. indicates the reference or comparison category.

Most other factors had no or negligible relationships with gross correction. However, confidence about one's initial decision predicted 14% lower odds of correcting one's response ( $p < 0.001$ ), time spent reflecting — socially or in isolation — predicted 7% higher odds of correction ( $p < 0.01$ ).

## Differences Between Solitary and Social Reflection Conditions

Given the non-random dissent-based matching algorithm, one may wonder if chat participants were more likely to correct their decision for reasons that had little to do with the chat. However, Figure A6 shows that chat participants were probably not different from essay participants. Chat participants were 8% more likely to spend another minute reflecting ( $p < 0.001$ ) and were 19% more likely to have a better experience in the experiment ( $p < 0.01$ ). However, even these associations are compatible with advantages of thinking together compared to thinking alone. Regardless, the two groups were largely similar on age, education, and other factors that might be thought to correct initial decisions independently of the chat intervention.

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| Prior intervention             | Description                                                                                | Reported effect                                                |
|--------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Budescu and Maciejovsky (2005) | Market incentives and information pooling across 16 rounds of Wason-card trading.          | About 60% improvement in correct card choices.                 |
| Boissin et al. (2023)          | Explanation of the correct and typical incorrect responses after each belief-bias problem. | About 10% increase in correct answers.                         |
| Isler and Yilmaz (2022)        | Five-item reflection test with a belief-bias item; training and delay manipulations.       | Mixed or limited generalization across tasks.                  |
| Boissin et al. (2022)          | Feedback, explanations, training, and delayed responding on reasoning problems.            | Up to 50% more correct final responses in training conditions. |
| Sirota et al. (2023)           | Repeated reflection prompts, feedback, explanations, and performance incentives.           | 11% to 40% more correct final responses.                       |
| Kneer et al. (2021)            | Delay, incentives, explanations, and a cognitive-reflection prime on a Gettier case.       | No reliable improvement in reflective thinking.                |

